Multimodal Feature Interactive Learning for Few-Shot Hyperspectral Image Classification

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***Abstract*—Recently, auxiliary cross-scene information has been widely utilized to improve the hypersperctral image classification performance by knowledge transfer. However, recognition of different objects with the same semantic category is difficult when the object types in similar scenes are different or only limited similarity knowledge is provided. In this article, a multimodal feature interactive (MMFI) learning method is proposed based on both hyperspectral image (HSI) modality and textual modality to distinguish similar objects, which enhances the transfer capability by utilizing the semantic prior from the textual modality. First, the adversarial domain mapping (ADM) module is designed to realize cross-domain knowledge transfer across different scenes in an adversarial learning manner. In particular, the noise is simulated as data distribution in different domains through domain mapping and aggregated with source and target domain data, which is then reconstructed and optimized to learn discriminative and conducive information for transfer. Then, the adaptive interaction learning (AIL) module acts on the latent features of the encoder to mine latent associations among the aggregated features and facilitate the expression of consistent features. In addition, few-shot learning (FSL) with textual embedding enables more powerful semantic priors for few-shot prototypes, making up for insufficient recognition capability in the presence of HSI modality only. Experimental results on three datasets demonstrate the superiority of our method.**

***Index Terms*—Classification, few-shot learning (FSL), hyperspectral image (HSI), multimodal, transfer learning.**

I. Introduction

HYPERSPECTRAL image (HSI) has received a lot of attention in recent years due to its special imaging conditions [1], which provide much more information than ordinary images. This information can be applied to various fields [2], [3], [4], such as medical diagnosis [5], food safety [6], and aerospace [7]. HSI classification is one of the fundamental research directions in the relevant fields, and its methods aim at dividing pixels into specific classes. In the early stage of research, manual feature extraction [8] and traditional classifiers [9], such as K-nearest neighbor (KNN) [10], support vector machine (SVM) [11], and principal component analysis [12], are used for HSI classification.

In recent years, few-shot learning (FSL) [13] has attracted significant interest for its ability to classify new and unseen classes with only a few training samples. One of the learning strategies in FSL, metalearning [14], uses an episodic learning [15] method. In this training scheme, the dataset is decomposed into different metatasks to gain metaknowledge and learn the model's generalization ability for few-shot scenes. The classification can be achieved without changing the trained model during the testing stage. Thereafter, the application of FSL in HSI classification has made significant progress. Liu et al. [16] propose a metric learning method based on deep residual 3-D convolution to effectively alleviate the small sample problem. Zhang et al. [17] construct a global prototype network to enhance the generalization ability of the model by generating new samples. Li et al. [18] integrate active learning with prototype networks to actively select high-value sample annotations to further improve classification performance. Liu et al. [16] propose deep FSL (DFSL), using a deep residual 3-D CNN network to achieve good classification results with few labels. Peng et al. [19] propose a convolutional transformer-based FSL (CTFSL) framework that synergistically leverages metalearning, adversarial domain alignment, and a hybrid CNN and Vision Transformer feature extractor to achieve outstanding performance. However, a shared encoder for both source and target HSI domains typically suffers from model degradation due to distribution misalignment.

With the constraint of scarce annotations, transferring knowledge between the source domain and the target domain presents a greater challenge, referred to as the cross-domain FSL (CDFSL) task in HSIC [20], [21]. CDFSL aims to tackle the challenge where a model trained on a label-rich source domain is required to recognize new classes in a target domain with only a few labeled samples [22]. Typically, episodic metatraining is conducted on the source domain by constructing repeated N-way K-shot tasks [20], enabling the network to learn an embedding space that reduces within-class variance while enlarging interclass margins [23]. Once

Received 27 February 2025; revised 12 May 2025, 27 July 2025, and 11 October 2025; accepted 17 October 2025. Date of publication 23 October 2025; date of current version 12 November 2025. This work was supported in part by the National Nature Science Foundations of China under Grant 62471236 and Grant 62471235, and in part by the Frontier Technologies Research and Development Program of Jiangsu under Grant BF2024070. *(Corresponding author: Fang Liu.)*

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Digital Object Identifier 10.1109/TGRS.2025.3624605

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such a discriminative metric space is established, domain discrepancy is mitigated through strategies such as adversarial domain confusion or alignment of task-relevant feature subspaces [24]. Wang et al. [25] propose a heterogeneous FSL framework to achieve cross-domain knowledge transfer from natural images to HSIs. In recent research, Liu et al. [26] propose to extract depth information from large models and fuse it with HSIs to improve the accuracy of small sample classification. It uses the sliding window method to extract the depth features of different bands, so as to obtain more rich depth features. Dong et al. [27] propose a deep metric learning method based on Brownian distance covariance for few-shot HSIC. They design a dual-channel Brownian distance covariance feature extraction network. Li et al. [28] propose SCFormer, a Transformer with spectral coordinate priors and two masking strategies for spectral feature discrimination. These strategies may not be effective across all datasets or applications. Xi et al. [29] introduce a hybrid model combining metalearning, CNNs, and Transformers with contrastive losses, pseudo-labeling, and data augmentation to improve feature discriminability. Its performance depends heavily on pseudo-label accuracy.

When utilizing cross-scene auxiliary information, the lack of an explicit semantic prior often results in misclassification of different objects within the same semantic category and similar objects across different categories. To enhance semantic discriminability in HSI classification, we propose the MMFI method, which incorporates textual modality as a complementary source to the HSI modality. MMFI includes three components: the adversarial domain mapping (ADM) module, the adaptive interaction learning (AIL) module, and FSL with textual embedding. The ADM module uses artificial noise to simulate source and target domain distributions, enhancing model generalization through adversarial learning. The AIL module refines latent features by improving the representation of aggregated information and reducing domain differences through adaptive interaction. The FSL component introduces textual embeddings to provide semantic priors, addressing the limitations of using only HSI data. These priors are treated as new prototypes for metric-based FSL in both domains, preserving transferable metaknowledge in the network. During testing, a KNN classifier is used to predict labels of unlabeled target samples. The main contributions of this article are shown below.

1) In transfer learning, both the source and target domains can be considered as a combination of common and discriminative information. However, directly utilizing discriminative information is challenging due to the difficulty of distinction. In this article, we successfully simulate the discriminative information of both domains through the ADM module, which not only fully exploits the source domain features, but also learns more useful knowledge for domain transfer to the target domain with adversarial learning.
2) To further explore the consistent representation between different aggregated latent features, we introduce the AIL module and utilize three parts to achieve adaptive interaction of the encoding representation. The process utilizes the idea of bridge to capture features that facilitate knowledge transfer.
3) In order to improve the ability to recognize different object types in similar scenes, we introduce the FSL with textual embedding. The textual modality enables to provide more powerful semantic prior for few-shot prototypes and makes up for the shortcomings when only the HSI modality is present.

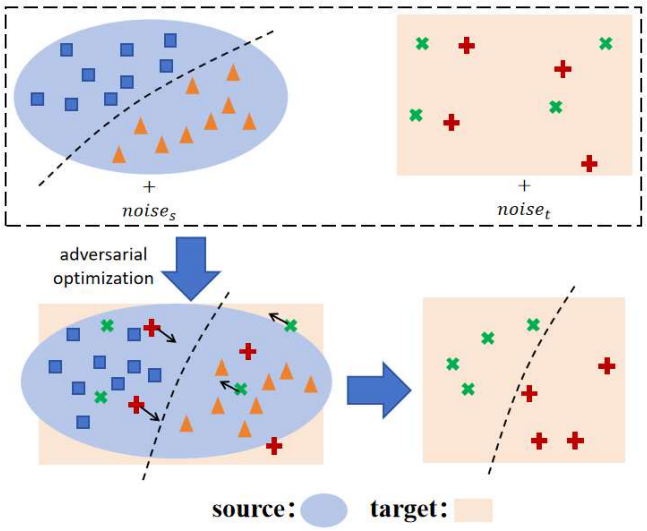


Fig. 1. Process of domain adaptation.

The remaining parts of this article are organized as follows. Section II describes the motivation of the proposed method. Section III describes the details of the MMFI method. Section IV shows the results of experiments and makes the analysis. Section V draws the conclusions according to the experimental results.

II. Motivation Description

Our motivation could be illustrated by Fig. 1. In order for the target domain to learn the data distribution of the source domain, we simulate noise and the data distribution in different domains through domain adaptation.

Consider two domains: the source domain and the target domain. The categories in the source domain have a clear distribution, while the categories in the target domain have a chaotic distribution. In order to make the target domain learn the data distribution of the source domain, noise_s and noise_t are simulated. Under the influence of noise, the data distribution of the source domain has also changed slightly, but it still maintains a clear distribution. Through adversarial optimization, the latent representations arising from target domain are progressively reshaped until their distribution coincides with that of source domain, thereby enabling robust cross-domain knowledge transfer and improve generalization on the target task.

In addition, when the same semantic categories exist in an HSI dataset, the recognition results for these categories may be inconsistent because they may differ significantly in their distributions within the feature space. For example,

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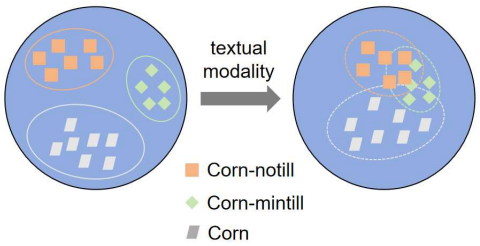


Fig. 2. Textual modality provides complementary information to HSI.

in Fig. 2, the blue area represents an HSI dataset, and the three different colors indicate different categories that belong to the same semantic type. Due to the lack of semantic prior knowledge, the use of category labels may result in a large difference between the recognition accuracy of one category and that of others, thus affecting the overall classification accuracy. In order to improve the consistency of recognizing different objects categories with the same or similar semantics, we introduce the textual modality to provide more powerful semantic information for HSI modality for FSL. When cross-scene HSI has dissimilar semantic categories or provides limited helpful knowledge, the textual modality is still able to play a role in the semantic enhancement of consistent representations of the same categories of objects.

III. Proposed Method

As shown in Fig. 3, there are three key parts in the proposed method, i.e., the ADM module, the AIL module, and the FSL with textual embedding.

For both source domain D_s and target domain D_t, the input is divided into the same size for network training. Therefore, we obtain source domain patch samples $p_s^0 \in R^{S\times S\times \mathrm{Chl}}$ and target domain patch samples $p_t^0 \in R^{S\times S\times \mathrm{Chl}}$ through the mapping layer, where the spatial size is $S \times S$ and the number of channels is Chl. For the two cross-scene datasets with domain shifts, we aim to utilize the information provided by the textual modality to compensate for the lack of semantic prior as well as to improve ability to recognize different objects.

For the ADM module, two domain mapping parts are designed to generate pseudo-distributions of the source and target domains, respectively. The generated simulated data are then combined with the source and target data to produce augmented representations of both domains as input to the network. Through the supervised training process, domain-specific knowledge is retained in the parameters of the domain mapping part. The encoder plays an important role in determining the effectiveness of knowledge transfer in cross-domain learning. In our approach, a deep 3-D residual network is employed as the encoder, which maps the spatial–spectral discriminative features of individual input data into the same metric space. The data distributions in different domains are aligned in an adversarial learning manner.

Then, the AIL module acts on aggregated latent features to enhance the consistent representation between different features through adaptive interaction. In addition, to obtain more complete semantic information about the two domains, an FSL with textual embedding utilizes textual modality to generate new prototypes. For both source and target domains, the new embedded features obtained by multimodal learning that adaptively combine textual modality and HSI modality features enable to represent the respective domains better. Then, new class prototypes computed from these features are used in FSL. The decoder improves the ability of the model to extract invariant representations by recovering the different latent representations as their respective original input images. The shared weights of the decoder ensure the presence of common information in the feature space.

After the feature extraction above, these features are input into two classifiers, the domain classifier and the few-shot classifier. The domain classifier's task is to determine whether the input sample belongs to the source or target domain, and it calculates a domain classification loss L_{dom} to aid in domain alignment. The few-shot classifier focuses on learning the characteristics of the few available samples from the target domain, generating a few-shot classification loss L_{fsl} to enhance performance in the target domain.

Finally, the framework combines the losses L_{gnn} which are generated by the AIL module, L_{dom} and L_{fsl} for joint optimization, enabling the model to effectively transfer knowledge between the source and target domains while achieving strong performance under few-shot conditions.

A. Adversarial Domain Mapping

In order to explore the discriminative features of the respective domains, we simulate the distributions of the source and target domains by transforming the random noise through the ADM module, thus building the prerequisites for direct manipulation of the simulated information to achieve domain adaptation. The ADM module is in front of the feature encoder and it consists of an upsampling function. The initialized random noises in the source and target domains follow the Gaussian distribution and is denoted as $N^0 \in R^{5\times5\times25}$. The dimension of the noise after domain mapping is consistent with p^0. The domain mapping function DM$(\cdot)$ learns different parameters θ_s and θ_t in the noise mapping process according to different source and target domains. Then, the obtained source noise and p_s^0, and the obtained target noise and p_t^0 are summed respectively to obtain the enhanced source and target domain representations, which are represented by P_s and P_t, respectively. In addition, the target noise and p_s^0 are also summed and represented by P_{trans} to learn the similar knowledge between the source and target domains, thus completing the knowledge transfer. The formulation of this process can be expressed as follows"

$$P_s = \mathrm{DM}\left(N_s^0;\theta_s\right) \oplus p_s^0 \tag{1}$$

$$P_t = \mathrm{DM}\left(N_t^0;\theta_t\right) \oplus p_t^0 \tag{2}$$

$$P_{\mathrm{trans}} = \mathrm{DM}\left(N_t^0;\theta_t\right) \oplus p_s^0. \tag{3}$$

These three types of samples are put into the encoder–decoder part, where the latent features and the corresponding global feature generated from the encoder $E(\cdot)$ are denoted by z_s, z_s^{glo}, z_t, z_t^{glo}, z_{trans}, and $z_{\mathrm{trans}}^{\mathrm{glo}}$,

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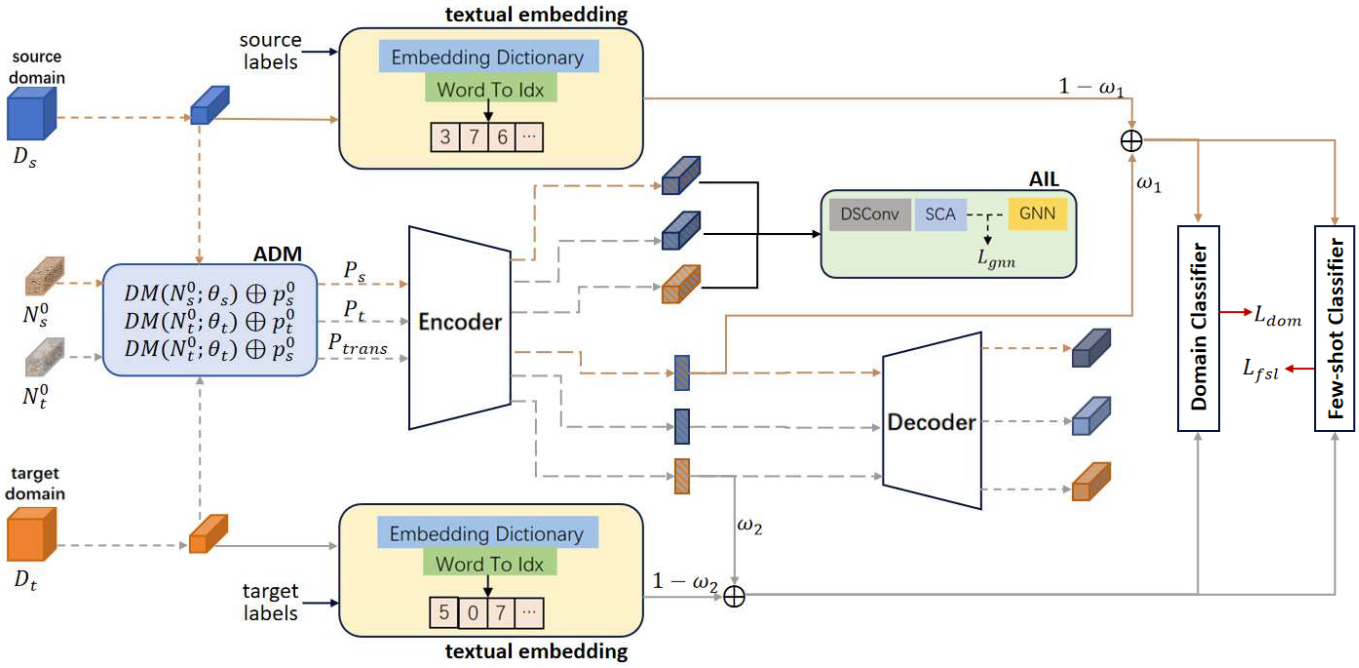


Fig. 3. Framework diagram of the proposed MMFI method. The ADM module simulates domain data distributions via noise and employs adversarial learning to enhance source and target domain representations, which are represented by P_s and P_t passed to the encoder. The AIL module learns consistent representations between different features through adaptive interaction. The method adaptively combines the HSI modality with the textual modality provided by the textual embedding to enhance the ability to recognize different categories. With the help of adaptive weights, the corresponding loss is obtained through domain classifier and few-shot classifier. Finally, the framework combines the losses for joint optimization.

respectively. The latent features are reconstructed by the feature decoder $D(\cdot)$ which has a symmetric structure with the encoder. In this process, the reconstruction operation enhances the representation of latent features and produces the reconstruction loss L_{recon}. The computational formula can be expressed as follows:

$$\tilde{P}_s = D\left(E\left(P_s\right)\right) \tag{4}$$

$$\tilde{P}_t = D\left(E\left(P_t\right)\right) \tag{5}$$

$$\tilde{P}_{\text{trans}} = D\left(E\left(P_{\text{trans}}\right)\right) \tag{6}$$

$$L_{\text{recon}} = \sum_{P_s,P_t,P_{\text{trans}}} \|\tilde{P} - P\|_2^2 \tag{7}$$

where $\tilde{P}_s$, $\tilde{P}_t$, and $\tilde{P}_{\text{trans}}$ denote the reconstructed samples, respectively.

The adversarial strategy is employed here to help the module learn domain information, as listed in (8). The discriminator Dis$(\cdot)$ with multilayer perceptron is utilized to distinguish data from different domains. In this way, the ADM module would be able to capture discriminative information related to domains. Therefore, the transformed samples P_{trans} contain specific domain information from the target domain and could be regarded as a bridge to learn transferable knowledge from the source domain. The formulation can be expressed as follows:

$$\begin{aligned}\max_{\text{Dis}} \min_{E} L_{\text{dom}} = {} & \mathbb{E}_{P_s\sim\hat{p}_s} \log\left(\text{Dis}\left(E\left(P_s\right)\right)\right) \\ & + \mathbb{E}_{P_t\sim\hat{p}_t} \log\left(1-\text{Dis}\left(E\left(P_t\right)\right)\right)\end{aligned} \tag{8}$$

where $\max_{\text{Dis}} \min_E L_{\text{dom}}$ represents an adversarial optimization process. The encoder E minimizes L_{dom} with the goal of learning domain-invariant features that can fool the Discriminator Dis. The discriminator Dis tries to maximize L_{dom}, accurately distinguishing the domain origin of the features to challenge the encoder. The ultimate optimization goal is to make the value of L_{dom} as low as possible, which indicates that the features extracted by the encoder have successfully achieved domain invariance and enabled cross-domain transfer. $\hat{p}_s$ and $\hat{p}_t$ denote the data distributions of P_s and P_t, respectively, and L_{dom} denotes the domain classification loss.

In order to ensure the semantic consistency of the reconstructed transformed data as well as the reconstructed source domain data and labels, the classification outputs of both and the labels use cross-entropy to calculate the loss, which can be expressed as follows:

$$L_{\text{trans}} = \text{CE}\left(\bar{P}_s, \bar{y}_s\right) + \text{CE}\left(\bar{P}_{\text{trans}}, \bar{y}_s\right) \tag{9}$$

where $\bar{P}_s$ and $\bar{P}_{\text{trans}}$ denote the representations of softmax classifier of $\tilde{P}_s$ and $\tilde{P}_{\text{trans}}$ respectively, $\bar{y}_s$ denotes one-hot labels of the source domain, CE$(\cdot,\cdot)$ denotes the cross-entropy function, and L_{trans} denotes the domain transfer loss.

Furthermore, we make the intermediate latent features maintain style consistency with the original source and target domains, yielding the style loss L_{style}. The formulation can be expressed as follows:

$$L_{\text{style}} = \sum_{z_s,z_t} \|z - z^0\|_2^2 \tag{10}$$

where z^0 denotes the latent features of the original data p^0 without combining noise.

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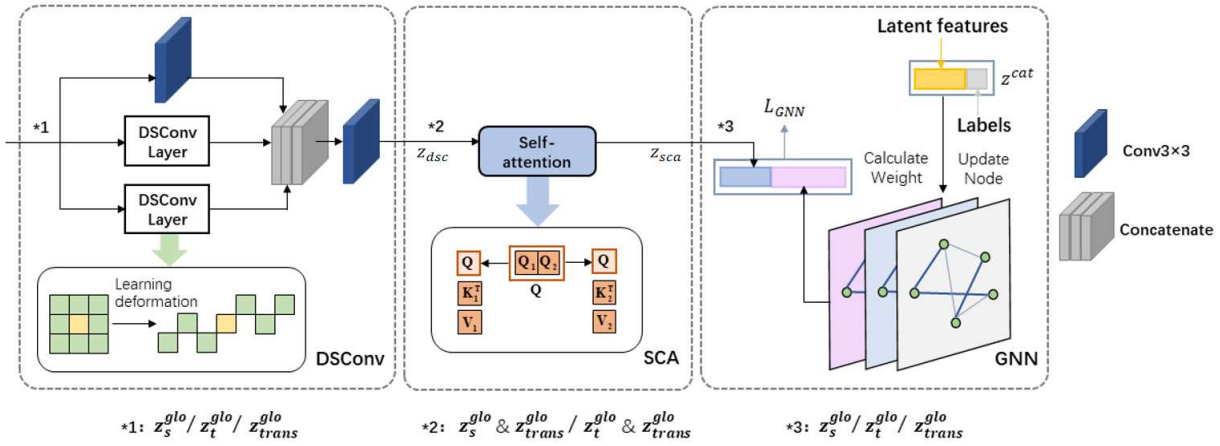


Fig. 4. AIL module is mainly composed of DSConv, SCA, and GNN parts. First, each of these three features passes through the DSConv part to realize the capture of different spatial patterns of the features. Then, these features are grouped and $z_{\text{trans}}^{\text{glo}}$ is used as a bridge to realize the adaptive interaction between different features. Finally, GNN is utilized to nearly mine the relationship between different features, thus facilitating the expression of consistent features.

The total loss L_{adm} of the adversarial domain mapping module is a linear combination of L_{recon}, L_{dom}, L_{trans}, and L_{style} with coefficients 2, 1, 2, 1, respectively,

$$L_{\text{adm}} = 2L_{\text{recon}} + L_{\text{dom}} + 2L_{\text{trans}} + L_{\text{style}}. \tag{11}$$

After the above operation, the ADM module can bridge the source and target domains by extracting domain-invariant features, enabling effective knowledge transfer while preserving semantic information.

B. Adaptive Interactive Learning

To fully leverage the intricate relationships among source domain features, target domain features, and the transformed features within the feature space, we propose the AIL module. This module is meticulously designed to enhance feature interaction and representation learning through three key components: the dynamic snake convolution (DSConv) which adaptively models complex spatial patterns, the spatial cross attention (SCA) which captures and aligns cross-domain spatial dependencies, and the graph neural network (GNN) which further refines and propagates feature relationships in a structured graph-based manner. These components synergistically enable the AIL module to extract richer, more transferable features across domains, facilitating robust domain adaptation.

The data processed by the ADM module is first fed into the DSConv component, which captures diverse spatial structures. Subsequently, the output is passed to the SCA module, where the attention mechanism is employed to strengthen the relationship between features from the source and target domains. Finally, the data is input into the GNN module to explore interfeature relationships and ensure consistency across domains. The structure of the AIL module is shown in Fig. 4, with the DSConv part, the SCA part, and the GNN part in order from left to right.

1) DSConv Part: For latent global feature z^{glo} in ADM module, it is captured by the encoder and is able to represent high-level semantic information of global range of pixels. However, the lack of capturing spatial dimension features may inhibit the discriminative spatial representation of different objects. Inspired by [30], the DSConv part is introduced to focus on the spatially complex and variable patterns of global features, hence it is able to quickly generalize when facing different patterns of features. This is useful for exploring the unity representation among the different pattern features of the three groups of features.

First, the source domain global features z_s^{glo}, the target domain global features z_t^{glo}, and the transformed data global features $z_{\text{trans}}^{\text{glo}}$ are input into the DSConv part, and each input is passed through three parallel branches, a 3-D convolutional layer and two dynamic snake convolutional layers, respectively, to obtain the output features of each branch. The aggregated features by channel are then passed through a 3-D convolutional layer with the number of channels restored to the preaggregation dimensions. The output feature z_{dsc} and input feature dimensions of the DSConv part are kept the same.

The dynamic snake convolutional layer is implemented by deforming the convolutional kernel so that it learns from the patterns of the features in the space and is implemented with the assurance that it does not drift away from the target. Thus, the selection of the convolution kernel for each step is based on the results of the previous step, and the position of the next one is decided based on the offset and interpolation. Therefore, a pixel site in the 2-D space of its input z^{glo} is randomly selected as the start point, and then moves within a distance of $[0, 4]$ from the center point with a bias of step size 1. The formulation for position selection can be expressed as follows:

$$\text{POS} = \sum_{\text{POS}'} B\left(\text{POS}', \text{POS}\right) \cdot \text{POS}' \tag{12}$$

$$B\left(\text{POS}, \text{POS}'\right) = b\left(\text{POS}_x, \text{POS}'_x\right) \cdot b\left(\text{POS}_y, \text{POS}'_y\right) \tag{13}$$

where $B(\cdot, \cdot)$ denotes the linear interpolation kernel, which consists of two 1-D kernels in the x-axis and y-axis directions, POS denotes the selected position, and POS′ denotes biasable positions. The interpolation is computed and summed over all biasable positions to obtain a new convolution kernel bias position adapted to the current features.

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2) SCA Part: The output of the DSConv part z_{dsc} is the input of the SCA part. First, the three different inputs are divided into two groups, the first group is the source domain data global feature z_s^{glo} and the transform data global feature $z_{\text{trans}}^{\text{glo}}$, and the second group is the transform data global feature $z_{\text{trans}}^{\text{glo}}$ and the target domain data global feature z_t^{glo}. $z_{\text{trans}}^{\text{glo}}$ is used as an intermediate feature to transfer the shared knowledge between different domains, thus realizing the interaction between different domain features. Then, the z_s^{glo} and $z_{\text{trans}}^{\text{glo}}$ are interacted using the attention mechanism. Both are mapped onto Q, K and V matrices by three 1×1 convolutional layers to obtain Q_1, Q_2, K_1, K_2, V_1, and V_2, respectively. Then, the concatenated matrix of Q_1 and Q_2 is used as the common query matrix for the two features. The new query matrix Q' is sequentially multiplied with the queried matrix K, which is then softmax normalized and multiplied with V to obtain the correlation representation between the two vectors. In addition, the computed correlation values are summed with the inputs to realize the capture of cross-data and cross-spatial dependencies. The formulation can be expressed as follows:

$$Q, K, V = \text{proj}\left(z_{\text{dsc}}\right) \tag{14}$$

$$Q' = \text{cat}\left(Q_1, Q_2\right) \tag{15}$$

$$z_{\text{sca}}^{\text{glo}} = z_{\text{dsc}} + \text{softmax}\left(\frac{Q'K^{\top}}{\sqrt{d_k}}\right)V \tag{16}$$

where proj$(\cdot)$ denotes the projection function, cat$(\cdot,\cdot)$ denotes the concatenate operation, softmax$(\cdot)$ denotes the softmax function, d_k denotes the dimension of the features, and $z_{\text{sca}}^{\text{glo}}$ denotes the output of self-supervised learning. The above operations apply to z_s^{glo}, $z_{\text{trans}}^{\text{glo}}$, and z_t^{glo}, which enrich the output transformed features with knowledge related to the source domain, and serve as an intermediate bridge to further improve the effectiveness of domain adaptation by enhancing the interactions between different features in an easily implementable way.

3) GNN Part: The GNN part is introduced to construct the topology to mine the inner connection between different features, so as to provide support for knowledge transfer. First, the source domain latent features, the target domain latent features, and the transformed latent features are concatenated with their respective labels to prepare for the graph initialization. An initialized graph structure is constructed based on attribute vectors such as vertices and edges, where each vertex z^{cat} is concatenated by latent features z_c' and labels y for each patch and the edge W is obtained by calculating the Euclidean distance between different vertices.

The update of the vertices is realized by multiplying and concatenating the edge weights with the concerned vertices. The computation process can be expressed as follows:

$$z_i^{\text{cat}} = \text{Update}\left(\text{cat}\left(W_{i,j} \cdot \text{reshape}\left(z_i^{\text{cat}}\right), z_i^{\text{cat}}\right)\right) \tag{17}$$

where Update$(\cdot)$ indicates the update operation for vertex i. This process does not change the connections between vertices and maintains the correlation between different features.

Next, the variant global features $z_{\text{sca}}^{\text{glo}}$ and the vertex z^{cat} that are updated via the graph structure are concatenated by channels. This operation fuses the interaction features acquired in the previous step to provide shared knowledge from other data for classification prediction. Moreover, the number of channels of global features and the latent features that are not updated by the graph structure remains the same. Finally, a fully connected layer is used to realize the category prediction and generate the loss L_{gnn} through the cross-entropy function, and the computation process can be expressed as follows:

$$L_{\text{gnn}} = \text{CE}\left(c_s^{\text{new}}, \bar{y}_s\right) + \text{CE}\left(c_{\text{trans}}^{\text{new}}, \bar{y}_s\right) + \text{CE}\left(c_t^{\text{new}}, \bar{y}_t\right) \tag{18}$$

where c_s^{new}, c_t^{new}, and $c_{\text{trans}}^{\text{new}}$ denote the category predicted by the fully connected layer for the source, target, and transformed features, respectively, and $\bar{y}_t$ denotes the one-hot labels of the target domain.

C. FSL With Textual Embedding

In order to solve the problem of limited information provided by only the HSI modality for the classification task, we introduce textual modality data as complements, which adaptively learns more complete and robust prior knowledge appropriate for the FSL by means of multimodal learning [31]. Therefore, the textual embedding is introduced in the multimodal learning process to implement enriched class prototypes. By establishing an embedding dictionary (ED) for all categories of each domain, samples with similar semantic labels are predicted to be of the same category, thus improving the model's ability to learn new categories. The structure of the textual embedding is shown in Fig. 5.

First, the source and target domain labels construct an ED based on the category names and the corresponding semantic categories. By finding the textual representations corresponding to the names of different sample categories, the categories can be semantically reified to obtain a new textual list. This list is then mapped to a collection of indexes, which can be easily processed by computer. Here, we introduce an unsupervised pretrained word embedding model M that contains embedding representations with respect to category c. For the input labels, the label embedding index list can be expressed as w_c. Then, the label embedding is transformed by $g(\cdot)$ to map the textual representation and the HSI representation to the same feature space, so that the dimensionality of the different representations is the same. The mapping process can be expressed as follows:

$$e_c = g\left(w_c\right) \tag{19}$$

where e_c denotes the output of the mapping process. In order to achieve consistency in multimodal feature space, we design a lightweight mapping function $g(\cdot)$ to map text representation or HSI features to a unified embedding space. The mapping function is implemented based on a linear transformation module consisting of 1×1 convolutions, and the specific structure is as follows:

$$g\left(x\right) = \text{BN}\left(\text{Conv2}D\left(x\right)\right). \tag{20}$$

Then, we assume that the multimodal features achieve feature combination according to the adaptive mixture coefficient

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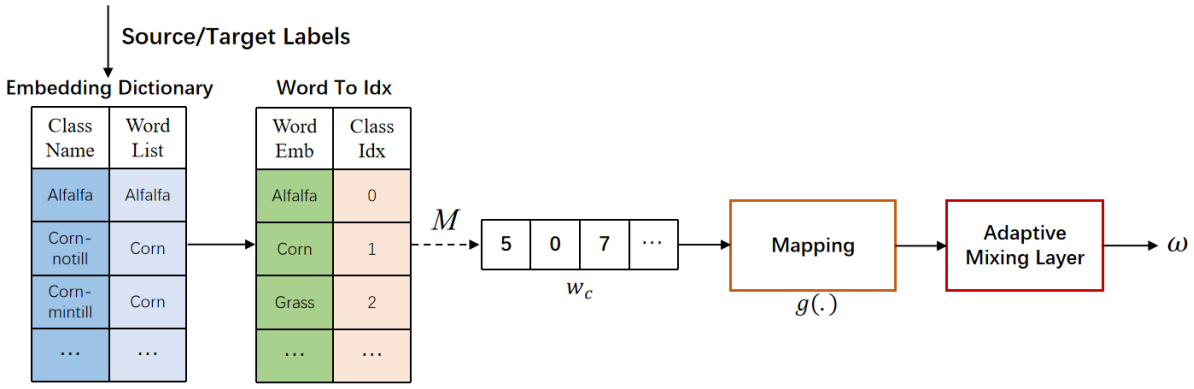


Fig. 5. Textual embedding. First, ED is constructed for the source and target domain labels, respectively. These labels are then leveraged, via the Word To Idx mapping and the matrix M, to subsequently update the textual embedding. The mapping operation $g(\cdot)$ then aligns the dimensions of the textual representation and the HSI representation. And adaptive mixing layer produces the coefficients of combining textual modality and HSI modality.

ω_c, which is related to the category, and the formulation can be expressed as follows:

$$\omega_c = \frac{1}{1+\exp\left(-\text{Adapt}\left(e_c\right)\right)} \tag{21}$$

$$z_c' = \omega_c \cdot z_c + (1-\omega_c)\cdot e_c \tag{22}$$

where z_c' and z_c denote the new latent features and the latent features obtained according to the category, respectively. Adapt$(\cdot)$ denotes the adaptive mixture transformation. The new latent features obtained are well suited to the new categories. We represent the multimodal features z_c' as the new prototype features.

For our approach, FSL is used to train the model to better transfer the source domain knowledge to the target domain. Specifically, a feature extractor is trained together using labeled samples from the source and target domains, and then generalization over the target domain is achieved. In FSL, the source and target domain samples involved in training are divided into different tasks according to episodes. The construction rule for each task is as follows: select c classes randomly from all the classes in the source domain, and select k labeled samples in each class to form a support set, denoted by S. Then, select n labeled samples randomly from the selected c classes to form a query set, denoted by Q. The task construction of the target domain dataset is similar to the source domain. The support and query sets of the source and target domains are denoted as S_s/Q_s and S_t/Q_t, respectively. Therefore, the division of training data is completed.

During the training process, the distribution of categories can be reflected by calculating the distance between the query sample and the prototype, which is calculated as follows:

$$p\left(y=c|P\in Q\right)\frac{\exp\left(-d\left(E\left(P\right),z_c'\right)\right)}{\sum_{k=1}\exp\left(-d\left(E\left(P\right),z_k'\right)\right)} \tag{23}$$

where z_c' denotes the prototype features of the cth class, which corresponds to the new features that adaptively combine the textual representations and HSI representations, and $d(\cdot,\cdot)$ denotes the Euclidean distance function [32]. Then, the FSL loss L_{fsl} could be expressed as follows:

$$L_{\text{fsl}} = \mathbb{E}_{S,Q}\left[-\sum_{(P,y)\in Q}\log p\left(y=c|P\right)\right]. \tag{24}$$

Therefore, the total loss can be expressed as a linear combination of the above losses, and the formulation can be expressed as follows:

$$\max_{\text{Dis}}\min_{E}\ L_{\text{adm}} + \sigma_1\cdot L_{\text{gnn}} + \sigma_2\cdot L_{\text{fsl}} \tag{25}$$

where σ_1 and σ_2 are parameters. By optimizing the ADM module loss L_{adm}, the AIL module loss L_{gnn}, and the few-shot classification loss L_{fsl}, the model converges and improves the expressiveness and generalization of the latent features. As a result, we can directly manipulate the optimized simulation information to achieve knowledge transfer across different domains and perform well on new unseen categories as well.

During the testing process, only the ADM module is involved in parameter updating, while the AIL module and the textual embedding keep the parameters of the training process.

IV. Experimental Results and Analysis

A. Datasets

We conduct experiments using the source domain and different target domains to demonstrate the effectiveness of the proposed method. In the experiments, Abidin et al. [33] is chosen as the source domain and the target domains are Indian pines (IPs) [34], Salinas (SAs) [35], and Center of Pavia (PC) [36], respectively. The source and target domain datasets are described in detail below.

1) Source Dataset: We chose Chikusei as the source dataset, and its image and category descriptions of are shown in Fig. 6 and Table I, respectively. The dataset is the image of the sky over the Japanese city of Chikusei taken by the Headwall Hyperspec-VNIR-C sensor at a wavelength of 343–1018 nm. The spatial size of the dataset is 2517 × 2335, the spatial resolution is 2.5 m, and the number of available wavebands is 128. There are 19 object classes in the research area, including both urban and rural areas.

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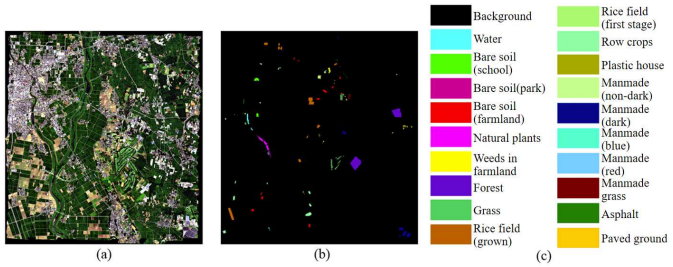


Fig. 6. Chikusei dataset. (a) Pseudo-color image. (b) Ground truth. (c) Sample categories.

TABLE I
DETAILS OF THE CHIKUSEI DATASET SAMPLE

Class	Name	Samples
1	Water	2845
2	Bare soil (school)	2859
3	Bare soil (park)	286
4	Bare soil (farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field (grown)	13369
10	Rice field (first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade (non-dark)	1220
14	Manmade (dark)	7664
15	Manmade (blue)	431
16	Manmade (red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145

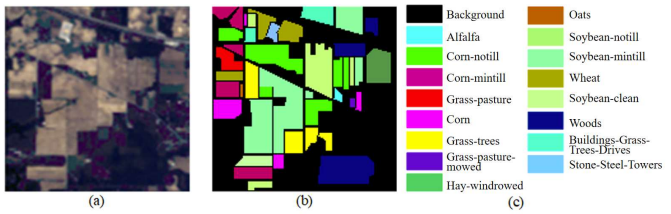


Fig. 7. IP dataset. (a) Pseudo-color image. (b) Ground truth. (c) Sample categories.

2) Target Datasets: The IP dataset is an image taken by the airborne visible/infrared imaging spectrometer (AVIRIS) over the state of Indiana, U.S.A., at wavelengths ranging from 400 to 2500 nm. Its spatial size is 145 × 145, spatial resolution is 20 m, and the number of available bands is 200. There are 16 object classes in the research area, including corn, oats, wheat, and woods. The dataset image and category descriptions are shown in Fig. 7 and Table II, respectively.

The SA dataset is an image taken by the AVIRIS sensor over Salinas Valley, California. Its spatial size is 512 × 217, spatial resolution is 3.7 m, and the number of bands actually used for training is 204. There are 16 object classes in the research area, including vegetables, bare soils, and vineyard fields. The dataset image and category descriptions are shown in Fig. 8 and Table III, respectively.

TABLE II
DETAILS OF THE IP DATASET SAMPLE

Class	Name	Samples
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93

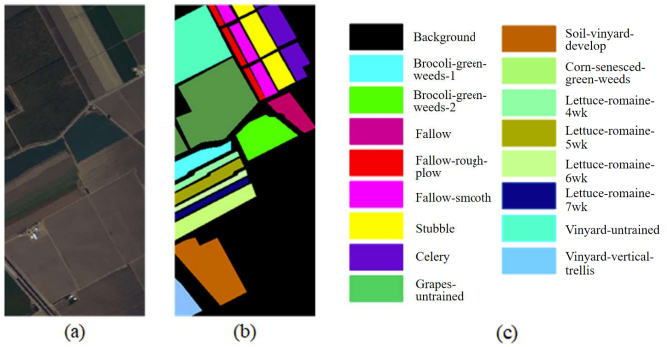


Fig. 8. SA dataset. (a) Pseudo-color image. (b) Ground truth. (c) Sample categories.

TABLE III
DETAILS OF THE SALINAS DATASET SAMPLE

Class	Name	Samples
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807

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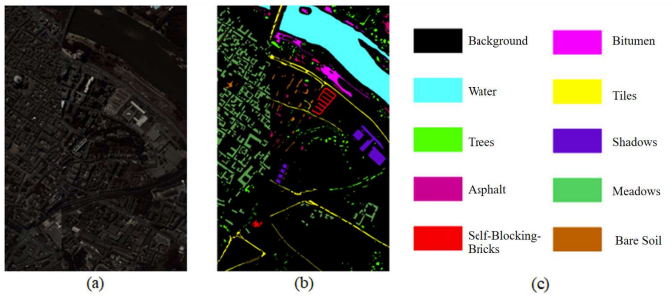


Fig. 9. PC dataset. (a) Pseudo-color image. (b) Ground truth. (c) Sample categories.

TABLE IV
DETAILS OF THE PC DATASET SAMPLE

Class	Name	Samples
1	Water	824
2	Trees	820
3	Asphalt	816
4	Self-Blocking Bricks	808
5	Bitumen	808
6	Tiles	1260
7	Shadows	476
8	Meadows	824
9	Bare Soil	820

The PC dataset is an image taken by the reflective optics system imaging spectrometer (ROSIS) sensor over Pavia, Northern Italy, in the wavelength range of 430–860 nm. Its spatial size is 610 × 340, spatial resolution is 1.3 m, and the number of bands actually used for the training is 102. There are nine object classes in the research area, including water, tree, shadow, and bare soil. The dataset image and category descriptions are shown in Fig. 9 and Table IV, respectively.

B. Experimental Settings

In this section, we choose Chikusei as the source domain and IP, SA, and PC as the target domains and conduct a series of experiments to verify the usefulness of our method. For the source domain data, each category with more than 200 samples is used for training. Since the FSL is used for training, for both source and target domains, c classes and k labeled samples from each class are selected randomly to form the training samples in each training task. Here k is set to 5 and c is set to 16, 16, and 9 in the IP, SA, and PC datasets, respectively. The rest of the target domain samples are used for the testing process. The size of the source and target domain data patches and the noise through domain mapping are both $9 \times 9 \times 100$. We set the number of iterations to 20 000 and select Adam optimizer for converging the network. In the total loss function, σ_1 and σ_2 are 1 and 7, respectively.

C. Experimental Results

In order to fully validate the effectiveness of the proposed method, two different types of experiments are designed: comparison and ablation experiments. To ensure fairness, the same hyperparameter settings are used for all experiments.

1) Comparison Experiment: In order to verify the advancement of the proposed method results, we compare it with other related methods as follows: the SSRN [37], DFSL+NN [16], DFSL+SVM [16], DCFSL [22], Gia-CFSL [38], CDSTN [39], HFSL [25], RPCL [40], SCFormer [28], and CTF-SSCL [29].

SSRN uses 3-D convolution to extract spatial–spectral discriminative features. DFSL+NN and DFSL+SVM are DFSL methods that utilize metalearning ideas to transfer knowledge from the source domain to the target domain. Gia-CFSL mainly improves the lack of domain alignment only for local spatial features by characterizing the nonlocal relationships within domains and capturing the interdomain similarities, and further enhances the cross-domain consistency representation by utilizing the graph structure alignment. CDSTN mainly designs a new feature extractor to capture the features and performs the domain alignment at different depths of the network while generating pseudo-labels for the target domains to improve the performance of the model. HFSL devises a spectral–spatial fusion network to make full use of the spatial and spectral information. RPCL proposes a refined prototypical contrastive learning network, which incorporates supervised contrastive learning and FSL into an end-to-end network. SCFormer and CTF-SSCL represent the latest methods in cross-domain few-shot HSI classification and are included in our experiments for comparison.

Our proposed method learns transferable source domain knowledge and improves the classification ability of the model on new categories by simulating the distribution of the respective domains and interacting with different features, and enriching the semantic features from the introduced textual modal data.

Tables V–VII show the results of the comparison experiments on the IP, SA, and PC datasets. The best results in each row are bolded. Analyzing the experimental results quantitatively and qualitatively separately, we can get some interesting findings. For the IP dataset, it can be seen from the classification metrics that our proposed method performs better than the other compared methods. Specifically, our method outperforms the best HFSL by 3.95%, 2.51%, and 4.36% for the three metrics of OA, AA, and kappa, respectively. It outperforms Gia-CFSL by 9.32%, 6.05%, and 10.43%, respectively. Compared to DCFSL, our method shows improvements of 10.92%, 8.03%, and 12.12% on the three metrics, respectively. In addition, our method compares with other traditional methods, deep learning methods, and FSL methods, the results have obvious advantages. On the SA dataset, compared with the best CDSTN, our method improves by 0.46%, 1.73%, and 0.94% on the three metrics of OA, AA, and kappa, respectively. Compared with DCFSL, our method improves by 3.94%, 1.05%, and 4.83% on the three metrics, respectively. Moreover, our method has higher accuracy values for the classification results than the other compared methods. For the PC dataset, our method outperforms other methods in classification metrics. Compared to the best result of CDSTN, our method improves by 0.12%, 1.20%, and 0.18% on the three metrics of OA, AA, and kappa, respectively. Moreover, the improvements of the proposed method over the HFSL method on the three metrics are 1.73%, 2.49%, and 2.42%,

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TABLE V
CLASSIFICATION RESULTS OF IP DATASET (%)

Class	SSRN [38]	DFSL+NN [19]	DFSL+SVM [19]	DCFSL [39]	Gia-CFSL [40]	CDSTN [41]	HFSL [42]	RPCL [43]	SCFormer [29]	CTF-SSCL [30]	Proposed
1	45.24	96.79	96.98	**100.00**	91.60	82.45	98.78	**100.00**	92.86	93.90	97.56
2	**88.04**	37.53	35.42	40.13	50.46	48.59	56.57	57.41	30.51	64.79	61.77
3	47.47	40.10	36.44	52.85	44.88	56.31	52.42	64.24	48.91	**69.21**	61.27
4	47.27	69.90	71.62	71.98	81.86	85.34	84.70	92.03	54.51	**92.89**	76.29
5	84.23	75.22	72.06	65.90	70.76	74.33	63.18	81.69	82.15	79.81	**85.25**
6	93.60	75.34	88.53	91.31	84.25	75.40	83.03	81.03	89.74	89.93	**96.14**
7	48.75	**100.00**	97.73	**100.00**	97.10	91.58	**100.00**	**100.00**	93.75	**100.00**	**100.00**
8	91.38	75.13	85.86	95.56	84.25	94.05	**99.47**	91.12	98.00	90.17	97.15
9	23.68	**100.00**	76.42	**100.00**	62.68	85.33	**100.00**	**100.00**	**100.00**	96.67	**100.00**
10	40.80	45.88	54.49	65.87	66.54	66.18	65.36	**70.37**	50.36	69.13	68.36
11	**73.22**	58.45	58.64	62.41	42.04	69.15	72.47	64.67	68.52	50.69	70.20
12	55.68	40.03	32.92	39.97	**97.11**	62.38	60.71	45.92	43.89	53.15	75.51
13	67.63	95.30	97.06	99.00	87.10	90.00	**99.50**	99.00	98.51	94.25	97.00
14	95.92	84.94	85.20	88.89	68.74	81.43	86.79	96.67	79.30	90.16	**96.87**
15	62.54	67.02	72.08	53.28	69.24	75.61	**98.16**	74.02	66.75	67.85	71.13
16	60.46	97.33	99.79	98.86	97.47	84.06	93.18	98.86	96.07	**100.00**	**100.00**
OA	62.76 ±0.35	60.05 ±0.79	61.34 ±1.88	65.80 ±2.76	67.40 ±0.60	69.19 ±4.85	72.77 ±0.39	72.46 ±0.11	63.98 ±0.52	70.20 ±4.57	**76.72 ±0.90**
AA	64.12 ±0.03	72.14 ±1.96	72.48 ±1.55	76.63 ±1.60	78.61 ±0.71	76.39 ±2.10	82.15 ±0.92	82.32 ±0.37	74.61 ±0.11	81.41 ±0.93	**84.66 ±1.00**
kappa	57.45 ±0.21	54.60 ±0.58	55.82 ±0.79	61.46 ±1.58	63.15 ±0.24	65.28 ±4.44	69.22 ±0.47	69.04 ±0.14	59.05 ±0.40	66.55 ±4.89	**73.58 ±0.95**

TABLE VI
CLASSIFICATION RESULTS OF SA DATASET (%)

Class	SSRN [37]	DFSL+NN [16]	DFSL+SVM [16]	DCFSL [22]	Gia-CFSL [38]	CDSTN [39]	HFSL [25]	RPCL [40]	SCFormer [28]	CTF-SSCL [29]	Proposed
1	97.55	95.63	73.92	98.24	99.88	97.80	99.55	98.45	94.86	97.38	**99.98**
2	98.97	99.09	96.85	99.33	99.34	98.15	98.13	99.74	98.35	**99.87**	99.86
3	92.47	94.01	96.28	91.75	95.04	88.91	92.77	88.13	**99.95**	96.09	90.56
4	96.50	99.54	99.11	98.59	96.99	99.60	**99.96**	99.53	99.21	97.88	76.29
5	94.20	90.58	80.72	95.01	89.02	94.23	**98.58**	91.30	95.89	89.04	87.71
6	99.28	98.47	91.63	**100.00**	99.52	97.22	99.06	98.94	99.68	96.85	**100.00**
7	**99.98**	99.81	97.73	98.12	99.44	94.93	98.34	99.86	99.64	99.37	99.48
8	86.90	77.74	82.33	60.65	80.89	**87.50**	71.50	76.66	69.00	72.47	83.66
9	99.64	91.13	94.44	99.88	99.55	97.09	98.62	99.92	**99.94**	97.68	99.59
10	**92.01**	60.98	80.96	89.63	76.47	85.08	79.22	86.85	84.97	79.39	85.65
11	95.56	95.99	93.38	98.73	97.74	91.38	98.92	98.97	92.86	93.23	**99.01**
12	99.15	93.13	97.94	99.13	98.39	96.45	95.14	**99.87**	96.23	97.74	99.04
13	89.24	99.34	95.79	98.59	98.87	96.35	98.85	**99.73**	99.62	99.18	98.90
14	95.15	98.06	98.87	99.71	95.69	98.71	98.83	**99.72**	98.55	96.06	98.01
15	55.97	77.54	71.73	73.05	75.76	77.60	75.51	**78.00**	59.13	83.08	76.35
16	98.91	85.05	90.57	**99.81**	94.08	87.19	98.03	84.96	99.08	89.65	97.97
OA	86.39 ±0.49	87.05 ±0.63	86.95 ±1.30	87.41 ±0.86	89.95 ±3.21	90.89 ±1.45	88.39 ±1.23	89.80 ±1.56	86.21 ±1.06	88.71 ±1.12	**91.35 ±1.09**
AA	93.24 ±0.20	91.01 ±0.42	90.08 ±1.44	93.69 ±0.42	93.74 ±0.48	93.01 ±0.95	93.81 ±1.52	93.79 ±0.56	92.93 ±0.50	92.81 ±0.92	**94.74 ±1.50**
kappa	84.95 ±0.48	85.63 ±0.52	85.51 ±1.42	85.94 ±0.95	88.82 ±1.24	89.83 ±1.61	87.10 ±1.41	88.67 ±1.70	84.67 ±1.16	87.46 ±1.25	**90.77 ±1.23**

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TABLE VII
CLASSIFICATION RESULTS OF PC DATASET (%)

Class	SSRN [37]	DFSL+NN [16]	DFSL+SVM [16]	DCFSL [22]	Gia-CFSL [38]	CDSTN [39]	HFSL [25]	RPCL [40]	SCFormer [28]	CTF-SSCL [29]	Proposed
1	99.82	99.10	99.05	99.45	99.62	99.79	99.24	99.01	99.82	99.19	**99.95**
2	93.66	89.93	88.24	90.49	90.63	93.10	90.36	94.15	**96.62**	95.13	94.62
3	80.48	93.03	93.40	**94.49**	94.10	87.86	73.84	92.17	78.37	88.64	91.94
4	76.52	83.73	85.14	**99.66**	93.09	93.51	99.33	73.27	90.32	91.31	96.96
5	95.31	80.83	83.76	85.54	88.14	94.33	**97.67**	97.35	90.21	87.44	94.89
6	84.24	90.83	92.28	95.06	94.73	96.50	86.08	92.87	97.56	92.99	**98.64**
7	**96.92**	83.25	84.96	83.14	84.42	83.87	94.41	86.26	84.68	84.37	83.14
8	98.30	97.71	97.01	98.66	98.24	98.81	95.16	98.38	**98.94**	98.45	97.73
9	98.02	95.73	96.20	95.14	98.33	97.78	97.88	96.90	**98.74**	95.92	98.51
OA	96.70 ±0.76	95.65 ±1.04	95.62 ±0.98	96.89 ±1.01	96.94 ±0.58	97.53 ±0.16	95.92 ±1.71	96.84 ±0.44	97.45 ±0.19	96.71 ±0.14	**97.65 ±0.30**
AA	91.67 ±0.37	90.46 ±2.52	90.38 ±2.05	93.51 ±0.42	93.48 ±0.84	93.95 ±0.80	92.66 ±0.69	92.26 ±0.04	92.81 ±0.67	92.61 ±0.34	**95.15 ±0.65**
kappa	94.84 ±1.02	93.87 ±1.26	93.96 ±0.79	95.60 ±0.18	95.68 ±0.82	96.51 ±0.23	94.27 ±2.37	95.54 ±0.61	96.39 ±0.27	95.34 ±0.19	**96.69 ±0.43**

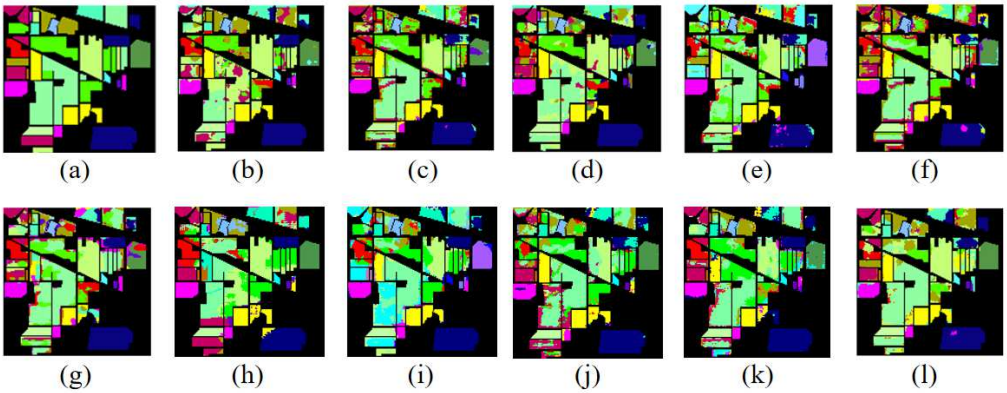


Fig. 10. Classification results for different methods on IP dataset. (a) Ground truth. (b) SSRN. (c) DFSL+NN. (d) DFSL+SVM. (e) DCFSL. (f) Gia-CFSL. (g) CDSTN. (h) HFSL (i) RPCL (j) SCFormer. (k) CTF-SSCL. (l) Our proposed method.

respectively. The other compared methods do not perform as well as our method in all three metrics, as reflected in the gain.

For the recent methods Gia-CFSL, CDSTN, RPCL, SCFormer, and CTF-SSCL, we have conducted a specific analysis. For the IP dataset, the proposed method improves by 9.32%, 6.05%, and 10.43% on OA, AA, and kappa compared with Gia-CFSL, improves by 7.53%, 8.27%, and 8.30% on OA, AA, and kappa compared with CDSTN, improves by 4.26%, 2.34%, and 4.54% on OA, AA, and kappa compared with RPCL, improves by 12.74%, 10.05%, and 14.53% on OA, AA, and kappa compared with SCFormer, and improves by 6.52%, 3.25%, and 7.03% on OA, AA, and kappa compared with CTF-SSCL. For the SA dataset, the proposed method improves by 1.40%, 1.00%, and 1.95% on OA, AA, and kappa compared with Gia-CFSL, improves by 0.46%, 1.73%, and 0.94% on OA, AA, and kappa compared with CDSTN, improves by 1.55%, 0.95%, and 2.10% on OA, AA, and kappa compared with RPCL, improves by 5.14%, 1.81%, and 6.10% on OA, AA, and kappa compared with SCFormer, and improves by 2.64%, 1.93%, and 3.31% on OA, AA, and kappa compared with CTF-SSCL. For the PC dataset, the proposed method improves by 0.71%, 1.67%, and 1.01% on OA, AA, and kappa compared with Gia-CFSL, improves by 0.12%, 1.20%, and 0.18% on OA, AA, and kappa compared with CDSTN, improves by 0.81%, 2.89%, and 1.15% on OA, AA, and kappa compared with RPCL, improves by 0.20%, 2.34%, and 0.30% on OA, AA, and kappa compared with SCFormer, and improves by 0.94%, 2.54%, and 1.35% on OA, AA, and kappa compared with CTF-SSCL. It can be seen that the proposed method has achieved all the best results in the three evaluation indicators. Also, the accuracy of each category shows the contribution of our method to different categories. It is worth mentioning that our method achieves more improvements on the IP dataset than the PC and SA datasets in terms of accuracy, i.e., the gain is higher. Figs. 10–12 show the visual classification results on the IP, SA, and PC datasets, respectively. Their use of different colors to indicate different categories of prediction results makes the results of the comparison experiments more visible.

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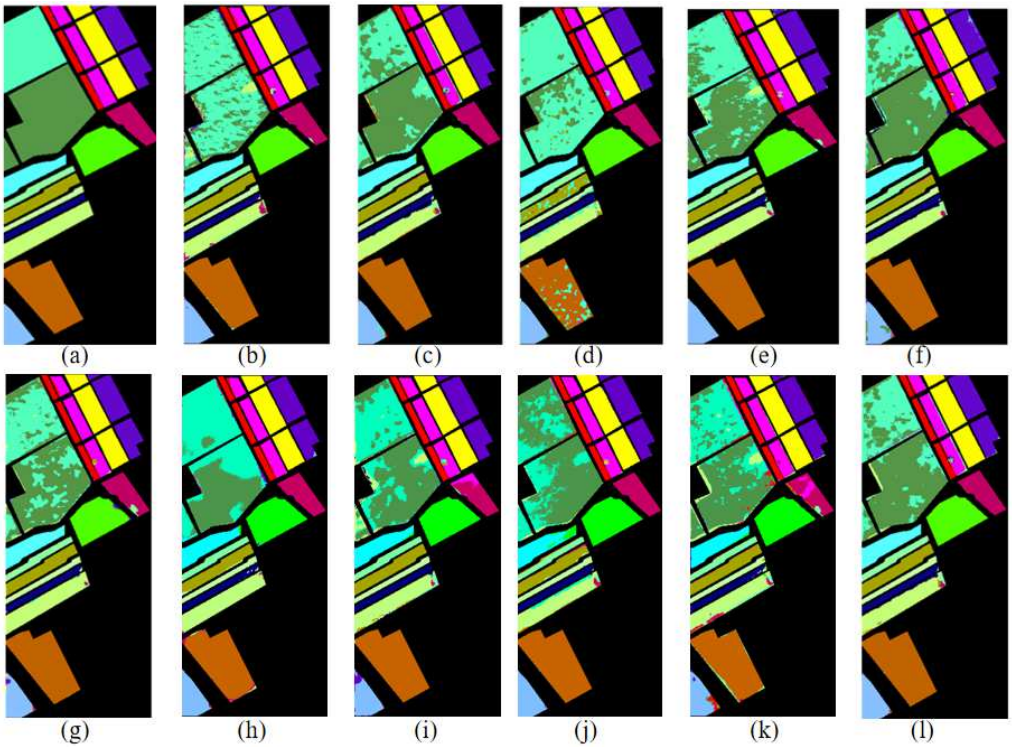


Fig. 11. Classification results for different methods on SA dataset. (a) Ground truth. (b) SSRN. (c) DFSL+NN. (d) DFSL+SVM. (e) DCFSL. (f) Gia-CFSL. (g) CDSTN. (h) HFSL (i) RPCL (j) SCFormer. (k) CTF-SSCL. (l) Our proposed method.

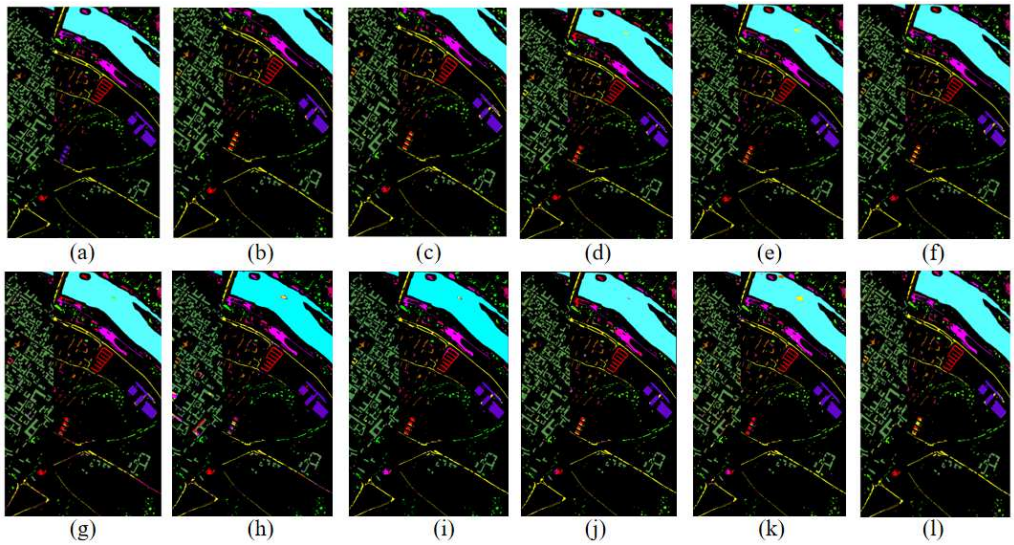


Fig. 12. Classification results for different methods on PC dataset. (a) Ground truth. (b) SSRN. (c) DFSL+NN. (d) DFSL+SVM. (e) DCFSL. (f) Gia-CFSL. (g) CDSTN. (h) HFSL (i) RPCL (j) SCFormer. (k) CTF-SSCL. (l) Our proposed method.

The comparison of the experimental results allows us to illustrate the differences between the different solution methods. The residual structure of SSRN enhances the feature extraction capability of the network. DFSL+NN and DFSL+SVM, based on DFSL method, address the generalization ability to different classes and overall images under small sample problems by constructing FSL tasks. For different datasets, different choices of classifiers produce some differences in the final results of the classification task. DCFSL addresses the cross-domain problem by introducing domain adaptation into the FSL process, acquiring the domain common knowledge and transferring it to the target domain by means of domain alignment. As a result, the classification accuracy of this method is higher than the previously mentioned methods on all three datasets. However, considering only local relationships leads to the inability to align interdomain features accurately. The Gia-CFSL considers nonlocal correlations across different domains and combines

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the graph structure to achieve domain distribution alignment. The CDSTN method expands the amount of labeled data in the target domain for the sample refinement pseudo-labeling operation, which further improves the performance of the model in the target domain classification task. As a result, compared with deep learning methods, FSL methods, and cross-domain methods, it can be seen that the proposed method performs more consistently on the whole range of categories, and seldom shows great deviations in prediction accuracy for different categories.

Our method outperforms all other approaches, not only because the simulated information in the cross-domain guided learning effectively captures the distribution characteristics of each domain, but also because the interaction between different sources of latent features enhances the expression of consistent representations. Moreover, the auxiliary information from the textual modality improves semantic consistency among similar categories within the feature space. This superiority is demonstrated by the greater stability of our method across all categories, higher evaluation metrics on all datasets, and notably stronger performance on datasets containing semantically similar classes.

This is shown by the fact that, taking the IP dataset as an example, corn types include corn-notill (class 2), corn-mintill (class 3), and corn (class 4) and soybean types include soybean-notill (class 10), soybean-mintill (class 11), and soybean-clean (class 12); the recognition accuracy remains stable for each class in corn and soybean types, rather than having some high and some low recognition accuracy for different classes with similar semantics. While, as the PC dataset, each type may contain only one class, its metrics gain is not as outstanding as that of the IP and SA datasets.

2) Ablation Experiment: In order to validate the contribution of our proposed modules to the classification task, we set up a set of ablation experiments. Since our method is based on DFSL and introduces the ADM module, the AIL module, and the textual embedding, the ablation experiments are designed around these components. The specific experimental design is represented as follows.

1) *Baseline:* DFSL.
2) *Proposed1:* Baseline + ADM.
3) *Proposed2:* Baseline + ADM + textual embedding.
4) *Proposed3:* Baseline + ADM + textual embedding + AIL.

We choose DFSL as Baseline and verify the role of our proposed modules based on it. In Proposed1, DSFL is combined with the ADM module, which contains related losses, i.e., the reconstruction loss L_{recon}, the domain transfer loss L_{trans}, the style loss L_{style}, and the domain classification loss L_{dom}. Therefore, it is compared with Baseline to show the effectiveness of our proposed ADM module. Proposed2 adds the textual embedding to Proposed1; by comparing the experimental results with Proposed1, we can verify the contribution of the textual modality in improving the classification results. Proposed3 adds the AIL module to Proposed2, the experiment is also the proposed MMFI method. The results of comparison with Proposed2 demonstrate the effectiveness of the AIL module. And the comparison with Baseline also illustrates the enhancement of HSI classification task when all modules act simultaneously.

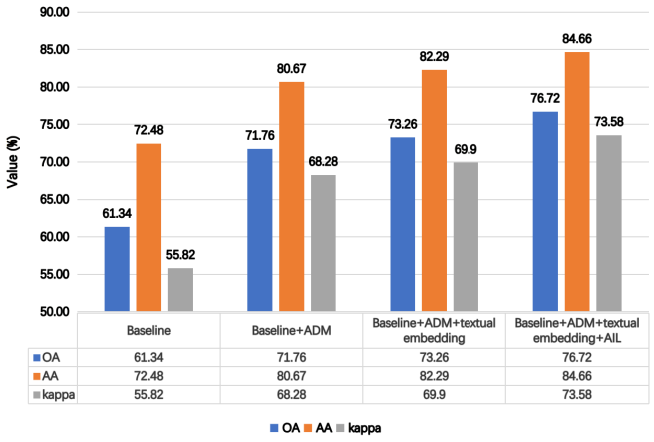


Fig. 13. Ablation experiment results on IP dataset.

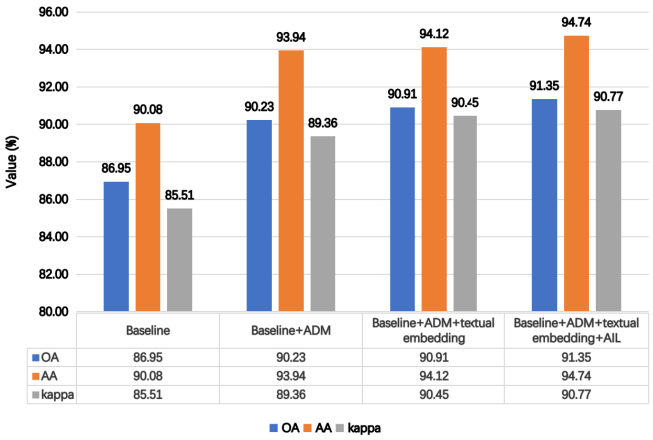


Fig. 14. Ablation experiment results on SA dataset.

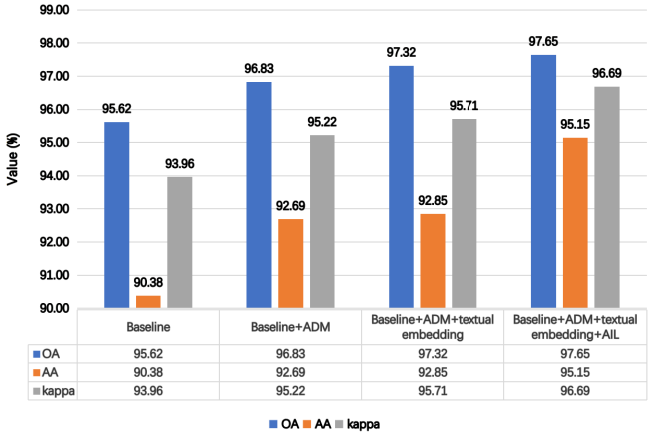


Fig. 15. Ablation experiment results on PC dataset.

We calculate the OA, AA, and kappa metrics for each experiment on the three datasets and compare the results as shown in Figs. 13–15. For the IP dataset, the ADM module improves by 10.42%, 8.19%, and 12.46% on OA, AA, and kappa, respectively, the textual embedding improves by1.50%,

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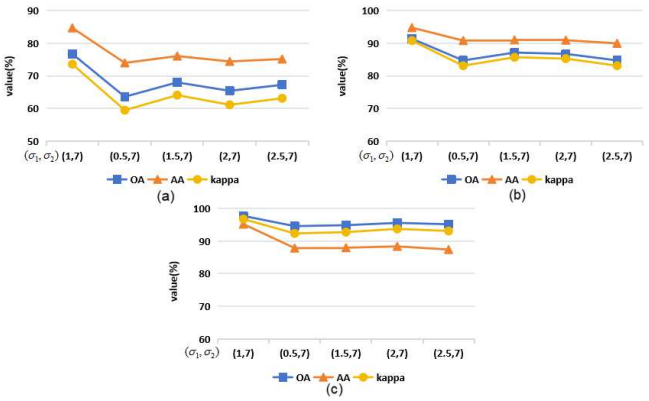


Fig. 16. Influence of the value of σ_1 on three datasets. (a) IP. (b) SA. (c) PC.

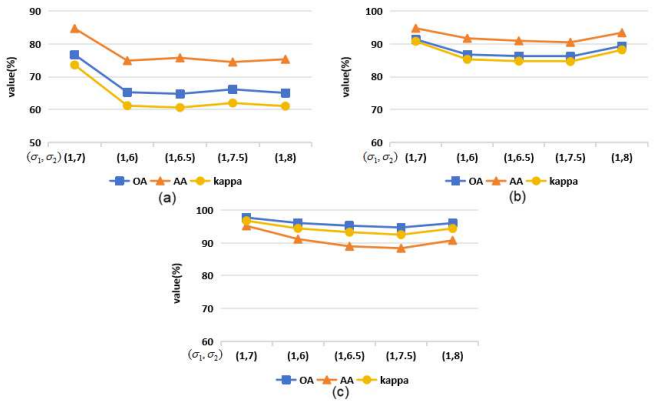


Fig. 17. Influence of the value of σ_2 on three datasets. (a) IP. (b) SA. (c) PC.

1.62%, and 1.62% on OA, AA, and kappa, respectively, and the AIL module improves by 3.46%, 2.37%, and 3.68% on OA, AA, and kappa, respectively. For the SA dataset, the ADM module increases by 3.28%, 3.86%, and 3.85% on OA, AA, and kappa, respectively, the textual embedding increases by 0.68%, 0.18%, and 1.09% on OA, AA, and kappa, respectively, and the AIL module increases by 0.44%, 0.62%,and 0.32% on OA, AA, and kappa, respectively. For the PC dataset, the ADM module provides improvements of 1.21%, 2.31%, and 1.26% on OA, AA, and kappa, respectively, the textual embedding provides improvements of 0.49%, 0.16%, and 0.49% on OA, AA, and kappa, respectively, and the AIL module provides improvements of 0.33%, 2.30%, and 0.98% on OA, AA, and kappa, respectively. Thus, all our proposed modules improve the classification performance on IP, SA, and PC datasets.

We compare the influence of the value of σ_1 in Fig. 16. It can be concluded that OA, AA, and kappa are the most effective when σ_1 is 1 and σ_2 is 7 on the three datasets. In addition, we compare the influence of the value of σ_2 in Fig. 17. It can be concluded that on the IP dataset, OA, AA, and kappa are the most effective when σ_1 is 1 and σ_2 is 7. On the SA and PC datasets, when σ_2 is 6, 6.5, or 7.5, the values of OA, AA, and kappa drop significantly. When σ_2 is 8, the AA value is slightly lower than when σ_2 is 7 on the SA dataset, but the OA and kappa values drop significantly. When σ_2 is 8, the OA and AA values are slightly lower than

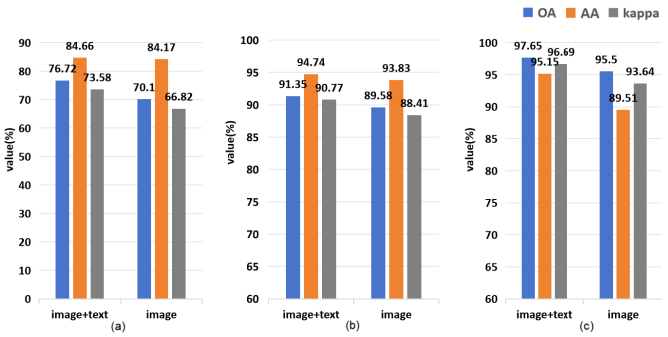


Fig. 18. Comparison of multimodal and unimodal results on three datasets. (a) IP. (b) SA. (c) PC.

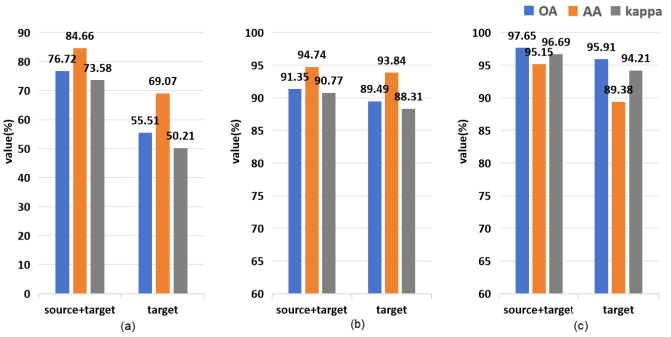


Fig. 19. Influence of FSL on three datasets. (a) IP. (b) SA. (c) PC.

when σ_2 is 7 on the PC dataset, but the kappa value drops significantly. Therefore, choosing σ_1 as 1 and σ_2 as 7 is the best option which shows the most balanced performance in the three indicators of OA, AA, and kappa, providing strong model stability while ensuring high accuracy. Therefore, it can be considered as the optimal parameter combination.

To verify the effectiveness of multimodal learning, Fig. 18 shows the comparison of results between multimodal and unimodal learning on the IP, SA, and PC datasets. For the IP dataset, the proposed multimodal learning method improves by 6.62%, 0.49%, and 6.76% on OA, AA, and kappa compared with unimodal learning, improves by 1.77%, 0.91%, and 2.36% on OA, AA, and kappa compared with unimodal learning on the SA dataset, and improves by 2.15%, 5.64%, and 3.05% on OA, AA, and kappa compared with unimodal learning on the PC dataset. It can be concluded that our proposed multimodal method performs significantly better than the unimodal approach. The unimodal learning performance is slightly inferior in all indicators.

To verify the effectiveness of few-shot learning, Fig. 19 explores the impact of FSL on the IP, SA, and PC datasets. Because FSL learns data transfer from the source, we removed the source for comparative experiments. From Fig. 19, it can be seen that the proposed FSL method improves by 21.21%, 15.59%, and 23.37% on OA, AA, and kappa on the IP dataset, improves by 1.86%, 0.90%, and 2.46% on OA, AA, and kappa on the SA dataset, and improves by 1.74%, 5.77%, and 2.48% on OA, AA, and kappa on the PC dataset. This reflects the advantages of FSL, while also indicating that the use of target data alone may lead to performance instability.

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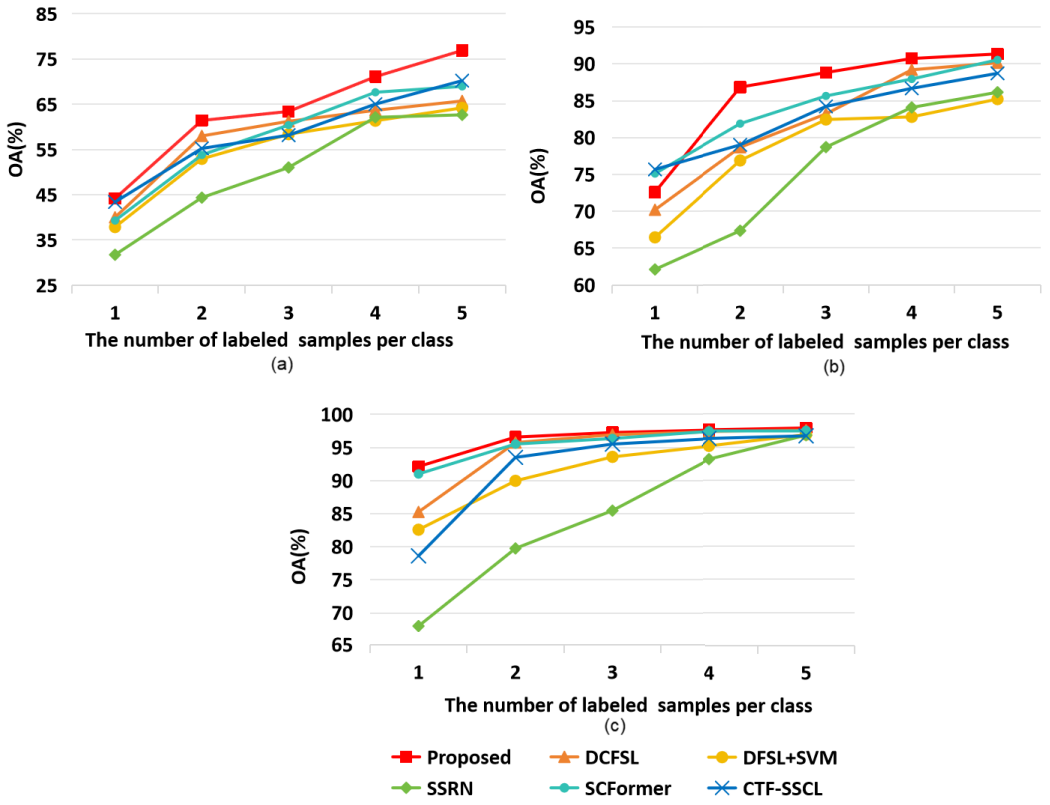


Fig. 20. Results for different numbers of labeled samples on three datasets. (a) IP. (b) SA. (c) PC.

3) Experiment About the Effect of Labeled Sample Size: In order to verify the robustness of our method on IP, SA, and PC datasets, we choose different number of labeled samples from each class to show the ability in maintaining the performance of the method in the face of fluctuations. We select 1, 2, 3, 4, and 5 labeled samples from each class of the target domain for few-shot training and compare results with SSRN, DFSL, DCFSL, SCFormer, and CTF-SSCL under the same parameters by calculating the OA, AA, and kappa metrics for each experiment. Fig. 20 describes the performance of the different methods with different number of labeled samples for the IP, SA, and PC, respectively.

Across all three datasets, the classification accuracy consistently improves with an increasing number of labeled samples. Among the compared methods, SSRN exhibits the most significant performance decline as the number of samples decreases. This is primarily because SSRN, as a deep learning model, requires a large number of labeled samples to effectively train its parameters. With limited data, its feature extraction capability is greatly weakened, resulting in lower accuracy and reduced robustness compared to FSL, cross-domain learning, and our proposed method. The DFSL method, which incorporates FSL, demonstrates stronger generalization under low-sample conditions and thus outperforms SSRN. DCFSL further improves performance by integrating both cross-domain learning and FSL, enabling effective knowledge transfer from source to target domains. As a result, it achieves higher accuracy than both SSRN and DFSL, and its performance degrades more slowly as the number of labeled samples decreases. On the IP and SA datasets, our proposed method significantly outperforms the two recent methods, SCFormer and CTF-SSCL. On the PC dataset, it achieves slightly better performance than SCFormer and notably better results than CTF-SSCL.

Overall, our method not only delivers the highest accuracy across all datasets but also exhibits the least performance fluctuation with varying numbers of labeled samples, indicating superior robustness. These results demonstrate that our approach is more robust and reliable compared to SSRN, DFSL, DCFSL, SCFormer, and CTF-SSCL.

D. Analysis of the Computational Complexity

Finally, we analyze the computational complexity of four algorithms: DCFSL, Gia-CFSL, SCFormer, and the proposed method, with the statistical results presented in Table VIII.

Among the four methods, Gia-CFSL exhibits the highest computational cost. Although it uses only 0.85M parameters, it incurs the highest computational cost, requiring 28.33G FLOPs on the IP and SA datasets and 9.07G on the PC dataset. Its inference times are also the longest, reaching 1.29, 11.56, and 18.38 s, respectively, indicating poor efficiency for practical use.

DCFSL uses 4.25M parameters and consumes 2.30G FLOPs on the IP and SA datasets and 1.53G on the PC dataset. However, its inference time remains relatively high, measured at 1.30 s on IP dataset, 11.41 s on SA dataset, and 15.97 s

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TABLE VIII
COMPARISON OF PARAMETERS, THE FLOPS, AND THE INFERENCE TIME OF THE PROPOSED METHOD AND THREE METHODS (DCFSL, GIA-CFSL, SCFORMER, AND CTF-SSCL)

		IP	SA	PC
DCFSL	para.		4.25M	
	FLOPs	2.30G	2.30G	1.53G
	time	1.30s	11.41s	15.97s
Gia-CFSL	para.		0.85M	
	FLOPs	28.33G	28.33G	9.07G
	time	1.29s	11.56s	18.38s
SCFormer	para.		7.14M	
	FLOPs	2.08G	2.08G	1.90G
	time	1.95s	19.94s	27.64s
Proposed	para.		0.90M	
	FLOPs	0.52G	0.52G	0.52G
	time	0.485s	0.485s	0.478s

on PC dataset. This indicates that despite moderate theoretical complexity, the model may suffer from practical inefficiencies during execution, possibly due to its parameter-intensive structure.

SCFormer, with 7.14M parameters, requires 2.08G FLOPs for IP and SA, and 1.90G for PC. Nonetheless, its inference time is the highest among all methods, reaching 1.95, 19.94, and 27.64 s, reflecting limited scalability and unsuitability for real-time tasks.

In contrast, the proposed method offers the most efficient profile, using only 0.90M parameters and 0.52G FLOPs across all datasets. It achieves the fastest inference times: 0.485 s on IP and SA, and 0.478 s on PC. These results clearly demonstrate the model's efficiency and practical deployment potential, especially in scenarios where computational resources and inference latency are critical factors.

Overall, the proposed method achieves the best trade-off between model size and speed, significantly outperforming the baselines in both theoretical and empirical efficiency.

V. CONCLUSION

In cross-domain learning, we propose a new HSI classification algorithm based on multimodal feature interactive (MMFI) learning by leveraging the semantic prior provided by textual modality for HSI modality. The ADM module is able to simulate the data distribution of the source and target domains, and the link between both domains is established directly by manipulating the simulated information. Then the module combines with adversarial learning to realize cross-domain knowledge transfer. Then, the AIL module, consisting of three parts, fully explores the relationships between different latent features through adaptive interactions, thus enhancing the consistent feature representation. In order to alleviate the limitation of absent semantic on HSI modality classification, the textual embedding provides textual modality for FSL. By adaptively combining textual and HSI modalities, it provides multimodal robust prior knowledge for learning while improving the consistent representation of category features in the presence of high semantic similarity. Multiple sets of experiments on three datasets also demonstrate the superiority of the proposed method on the HSI classification task.

Therefore, the proposed MMFI method and the proposed modules are effective in achieving knowledge transfer and improving the classification performance by utilizing the transferable knowledge. Although our method has excellent performance in knowledge transfer and classification in few-shot learning, the execution speed of the model is sometimes slow. Efforts will be made to improve the execution speed of the model.

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